

Introduction to Cryptography

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XOR outputs 1 when the bits differ

- XOR (exclusive OR), written $\oplus$, combines two bits using this truth table:

x	y	$x \oplus y$
0	0	0
0	1	1
1	0	1
1	1	0

- XOR is applied to bit strings one position at a time
- XORing with the same bit twice cancels it:

$$(x \oplus y) \oplus y = x.$$

From Vigenère to the one-time pad

- In the byte-wise Vigenère cipher, a short key is repeated across the message
 - Repetition creates patterns that an attacker can exploit.
- Instead, for an n -bit message m , choose a fresh, uniform n -bit key k and compute

$$c = m \oplus k.$$

- The receiver recovers the message because $c \oplus k = m$
- This scheme is the *one-time pad*: the key is as long as the message and must never be reused
- Later today, we will define perfect secrecy and prove that this scheme achieves it

Quiz

- What is the encryption of the plaintext 0101 0111 using the one-time pad with key 1111 1111?
 - 0101 0111
 - 1111 1111
 - 1010 1000
 - 0101 1010

Core principles of modern crypto

- Formal definitions
 - Precise, mathematical model and definition of what security means
- Assumptions
 - Clearly stated and unambiguous
- Proofs of security
 - Move away from design-break-patch

Importance of definition

Importance of definitions

- Definitions are *essential* for the design, analysis, and sound usage of crypto

Importance of definitions – design

- Developing a precise definition forces the designer to think about what they really want
 - What is essential and (sometimes more important) what is not
 - * Often reveals subtleties of the problem

Importance of definitions – design

- *If you don't understand what you want to achieve, how can you possibly know when (or if) you have achieved it?*

Importance of definitions – analysis

- Definitions enable meaningful analysis, evaluation, and comparison of schemes
 - Does a scheme satisfy the definition?
 - What definition does it satisfy?
 - * Note: there may be multiple meaningful definitions!
 - * One scheme may be less efficient than another, yet satisfy a stronger security definition

Importance of definitions – usage

- Definitions allow others to understand the security guarantees provided by a scheme
- Enables schemes to be used as *components* of a larger system (modularity)
- Enables one scheme to be substituted for another if they satisfy the same definition

Importance of assumption

Assumptions

- With few exceptions, cryptography currently requires *computational assumptions*
 - At least until we prove $P \neq NP$ (and even that would not be enough)
- Principle: any such assumptions should be made explicit

Importance of clear assumptions

- Allow researchers to (attempt to) *validate* assumptions by studying them
- Allow meaningful *comparison* between schemes based on different assumptions
 - Useful to understand minimal assumptions needed
- Practical implications if assumptions are wrong
- Enable proofs of security

Proofs of security

- Provide a rigorous proof that a construction satisfies a given definition under certain specified assumptions
 - Provides an iron-clad guarantee (relative to your definition and assumptions!)
- Proofs are crucial in cryptography, where there is a malicious attacker trying to “break” the scheme

Limitations?

- Cryptography remains partly an *art* as well
- Given a proof of security based on some assumption, we still need to *instantiate* the assumption
 - Validity of various assumptions is an active area of research

Limitations?

- Proofs given an iron-clad guarantee of security
 - ... relative to the definition and the assumptions!
- Provably secure schemes can be broken!
 - If the definition does not correspond to the real-world threat model
 - * I.e., if attacker can go “outside the security model”
 - * This happens a lot in practice
 - If the assumption is invalid
 - If the implementation is flawed
 - * This happens a lot in practice

Nevertheless. . .

- This does not detract from the importance of having formal definitions in place
- This does not detract from the importance of proofs of security

Defining secure encryption

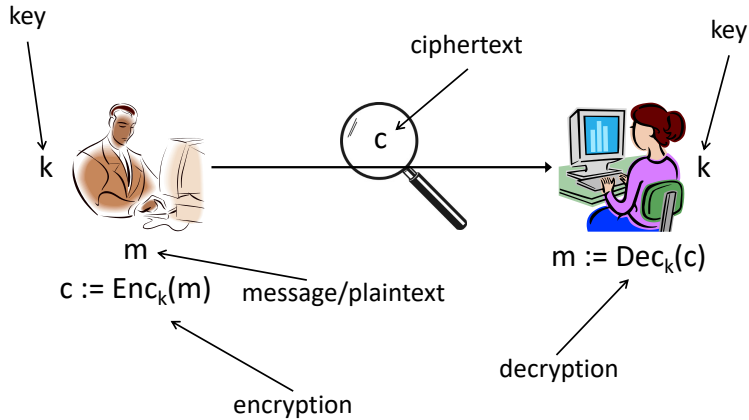
Crypto definitions (generally)

- Security guarantee/goal
 - What we want to achieve (or what we want to prevent the attacker from achieving)
- Threat model
 - What (real-world) capabilities the attacker is assumed to have

Recall

- A *private-key encryption scheme* is defined by a message space $\mathcal{M}$ and algorithms (Gen, Enc, Dec):
 - Gen (key-generation algorithm): generates k
 - Enc (encryption algorithm): takes key k and message $m \in \mathcal{M}$ as input; outputs ciphertext $c \leftarrow \text{Enc}_k(m)$
 - Dec (decryption algorithm): takes key k and ciphertext c as input; outputs $m := \text{Dec}_k(c)$

Private-key encryption



Threat models for encryption – some possibilities

- Ciphertext-only attack
 - One ciphertext or many?
- Known-plaintext attack
- Chosen-plaintext attack
- Chosen-ciphertext attack

Goal of secure encryption?

- How would you define what it means for encryption scheme (Gen, Enc, Dec) over message space $\mathcal{M}$ to be secure?
 - Against a (single) ciphertext-only attack

Secure encryption – key?

- “Impossible for the attacker to learn the key”
 - The key is a *means to an end*, not the end itself
 - Necessary (to some extent) but not sufficient
 - Easy to design an encryption scheme that hides the key completely, but is insecure
 - Can design schemes where most of the key is leaked, but the scheme is still secure

Secure encryption – plaintext/ciphertext?

- “Impossible for the attacker to learn the plaintext from the ciphertext”
 - What if the attacker learns 90% of the plaintext?
- “Impossible for the attacker to learn any character of the plaintext from the ciphertext”
 - What if the attacker is able to learn (other) partial information about the plaintext?
 - * E.g., salary is greater than \$75K
 - What if the attacker guesses a character correctly?

The right definition

- “Regardless of any *prior* information the attacker has about the plaintext, the ciphertext should leak no *additional* information about the plaintext”
 - How to formalize?

Perfect secrecy

Probability review

- *Random variable (r.v.):* variable that takes on (discrete) values with certain probabilities
- Probability distribution for a r.v. specifies the probabilities with which the variable takes on each possible value
 - Each probability must be between 0 and 1
 - The probabilities must sum to 1

Probability review

- *Event*: a particular occurrence in some experiment
 - $\Pr[E]$: probability of event E
- Conditional probability: probability that one event occurs, *given that* some other event occurred
 - $\Pr[A \mid B] = \frac{\Pr[A \cap B]}{\Pr[B]}$
- Two random variables X, Y are *independent* if, for all x, y ,

$$\Pr[X = x \mid Y = y] = \Pr[X = x].$$

Probability review

- Law of total probability: say $E_1, \dots, E_n$ are a *partition* of all possibilities. Then, for any event A ,

$$\Pr[A] = \sum_{i=1}^n \Pr[A \cap E_i] = \sum_{i=1}^n \Pr[A \mid E_i] \Pr[E_i].$$

Notation

- $\mathcal{K}$ (key space) – set of all possible keys
- $\mathcal{C}$ (ciphertext space) – set of all possible ciphertexts

Probability distributions

- Let M be the random variable denoting the value of the message
 - M ranges over $\mathcal{M}$
 - Context dependent!
 - Reflects the likelihood of different messages being sent, given the attacker's prior knowledge
 - E.g., $\Pr[M = \text{attack today}] = 0.7$ and $\Pr[M = \text{don't attack}] = 0.3$

Probability distributions

- Let K be a random variable denoting the key
 - K ranges over $\mathcal{K}$
- Fix some encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$
 - Gen defines a probability distribution for K :

$$\Pr[K = k] = \Pr[\text{Gen outputs } k].$$

Probability distributions

- Random variables M and K are *independent*
 - Require that parties don't pick the key based on the message, or the message based on the key

Probability distributions

- Fix some encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$, and some distribution for M
- Consider the following (randomized) experiment:
 - Generate a key k using Gen
 - Choose a message m according to the given distribution
 - Compute $c \leftarrow \text{Enc}_k(m)$
- This defines a distribution on the ciphertext!
- Let C be a random variable denoting the value of the ciphertext in this experiment

Example 1

- Consider the shift cipher
 - For every $k \in \{0, \dots, 25\}$, $\Pr[K = k] = 1/26$
- Say $\Pr[M = a] = 0.7$ and $\Pr[M = z] = 0.3$
 - Here $\mathcal{M} = \{a, z\}$
- What is $\Pr[C = b]$?
 - Either $M = a$ and $K = 1$, or $M = z$ and $K = 2$

$$\begin{aligned}\Pr[C = b] &= \Pr[M = 'a'] \Pr[K = 1] + \Pr[M = z] \Pr[K = 2] \\ &= 0.7 \left(\frac{1}{26} \right) + 0.3 \left(\frac{1}{26} \right) = \frac{1}{26}.\end{aligned}$$

Example 2

- Consider the shift cipher and the distribution on M given by

$$\Pr[M = \text{one}] = \frac{1}{2}, \quad \Pr[M = \text{ten}] = \frac{1}{2}.$$

- Here $\mathcal{M} = \{\text{one}, \text{ten}\}$
- What is $\Pr[C = \text{rqh}]$?
 - $\Pr[C = \text{rqh} \mid M = \text{one}] = 1/26$
 - $\Pr[C = \text{rqh} \mid M = \text{ten}] = 0$, i.e., no key maps `ten` to `rqh`

$$\Pr[C = \text{rqh}] = \frac{1}{26} \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = \frac{1}{52}.$$

Perfect secrecy (informal)

- “Regardless of any *prior* information the attacker has about the plaintext, the ciphertext should leak no *additional* information about the plaintext”

Perfect secrecy (informal)

- Attacker's information about the plaintext = attacker-known *distribution* of M
- Perfect secrecy means that observing the ciphertext should not change the attacker's knowledge about the distribution of M

Perfect secrecy (formal)

- Encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ and ciphertext space $\mathcal{C}$ is *perfectly secret* if, for every distribution over $\mathcal{M}$, every $m \in \mathcal{M}$, and every $c \in \mathcal{C}$ with $\Pr[C = c] > 0$,

$$\Pr[M = m \mid C = c] = \Pr[M = m].$$

- I.e., the distribution of M does not change conditioned on observing the ciphertext

Example 3

- Consider the shift cipher and the distribution

$$\Pr[M = \text{one}] = \frac{1}{2}, \quad \Pr[M = \text{ten}] = \frac{1}{2}.$$

- Take $m = \text{ten}$ and $c = \text{rqh}$
- $\Pr[M = \text{ten} \mid C = \text{rqh}] = 0 \neq \Pr[M = \text{ten}]$

Perfect secrecy is not satisfied!

Observing the ciphertext `rqh` changes the attacker's knowledge about the distribution of M

Bayes's theorem

- For events A and B with $\Pr[B] > 0$,

$$\Pr[A \mid B] = \frac{\Pr[B \mid A] \Pr[A]}{\Pr[B]}.$$

Example 4

- For the shift cipher, let

$$\Pr[M = \text{hi}] = 0.3, \quad \Pr[M = \text{no}] = 0.2, \quad \Pr[M = \text{in}] = 0.5.$$

- What is $\Pr[M = \text{hi} \mid C = \text{xy}]$?
 - By Bayes's theorem,

$$\Pr[M = \text{hi} \mid C = \text{xy}] = \frac{\Pr[C = \text{xy} \mid M = \text{hi}] \Pr[M = \text{hi}]}{\Pr[C = \text{xy}]}.$$

Example 4, continued

- $\Pr[C = xy \mid M = hi] = 1/26$
 - There is exactly one key that maps hi to xy, and all keys are equally likely
- By the law of total probability,

$$\begin{aligned}\Pr[C = xy] &= \Pr[C = xy \mid M = hi](0.3) \\ &\quad + \Pr[C = xy \mid M = no](0.2) \\ &\quad + \Pr[C = xy \mid M = in](0.5) \\ &= \frac{1}{26}(0.3) + \frac{1}{26}(0.2) + 0(0.5) = \frac{1}{52}.\end{aligned}$$

Example 4, continued

- What is $\Pr[M = \text{hi} \mid C = \text{xy}]$?
 - $\Pr[M = \text{hi} \mid C = \text{xy}] = \frac{(1/26)(0.3)}{1/52} = 0.6$
 - $0.6 \neq \Pr[M = \text{hi}] = 0.3$

Conclusion

- The shift cipher is not perfectly secret!
 - At least not for 2-character messages
- How to construct a perfectly secret scheme?

One-time pad

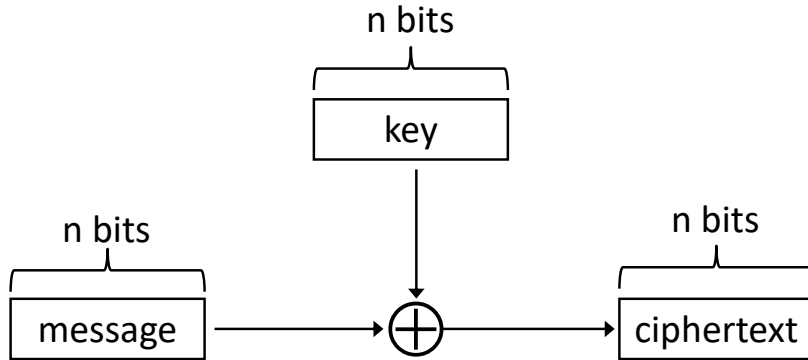
- Patented in 1917 by Vernam
 - Recent historical research indicates it was invented (at least) 35 years earlier
- Proven perfectly secret by Shannon (1949)

One-time pad

- Let $\mathcal{M} = \{0, 1\}^n$
- Gen: choose a uniform key $k \in \{0, 1\}^n$
- $\text{Enc}_k(m) = k \oplus m$
- $\text{Dec}_k(c) = k \oplus c$
- Correctness:

$$\text{Dec}_k(\text{Enc}_k(m)) = k \oplus (k \oplus m) = (k \oplus k) \oplus m = m.$$

One-time pad



Perfect secrecy of one-time pad

- Note that *any* observed ciphertext can correspond to *any* message (why?)
 - (This is necessary, but not sufficient, for perfect secrecy)
- So, having observed a ciphertext, the attacker cannot conclude for certain which message was sent

Perfect secrecy of one-time pad

- Fix an arbitrary distribution over $\mathcal{M} = \{0, 1\}^n$, and arbitrary $m, c \in \{0, 1\}^n$
 - We assume the key is chosen uniformly at random from $\{0, 1\}^n$
- By Bayes's theorem,
 - $\Pr[M = m \mid C = c] = \frac{\Pr[C = c \mid M = m] \Pr[M = m]}{\Pr[C = c]}$
- By the law of total probability,

$$\begin{aligned}\Pr[C = c] &= \sum_{m' \in \mathcal{M}} \Pr[C = c \mid M = m'] \Pr[M = m'] \\ &= \sum_{m' \in \mathcal{M}} \Pr[K = m' \oplus c \mid M = m'] \Pr[M = m'] \\ &= \sum_{m' \in \mathcal{M}} 2^{-n} \Pr[M = m'] = 2^{-n}.\end{aligned}$$

Perfect secrecy of one-time pad

- For arbitrary $m, c \in \{0, 1\}^n$,

$$\begin{aligned}\Pr[M = m \mid C = c] &= \frac{\Pr[C = c \mid M = m] \Pr[M = m]}{\Pr[C = c]} \\&= \frac{\Pr[K = m \oplus c \mid M = m] \Pr[M = m]}{2^{-n}} \\&= \frac{2^{-n} \Pr[M = m]}{2^{-n}} \\&= \Pr[M = m].\end{aligned}$$

- Therefore, the one-time pad is perfectly secret.